JavaTower — OOP Architecture Documentation

CIS096-1 — Principles of Programming and Data Structures
Assessment 2 — Week 9: OOP Architecture and Implementation

Field	Details
Unit Code	CIS096-1
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1. Introduction

JavaTower is a real-time tower defence RPG implemented in Java 21 using JavaFX for the graphical interface and SQLite (via JDBC) for persistent data storage. The game challenges the player to survive 30 waves of progressively more difficult undead enemies by controlling a hero character, placing defensive towers, managing an inventory of equipment, and upgrading skills.

Key Features

- **Real-time gameplay** with an `AnimationTimer` game loop running at 60 FPS
- **10-tier enemy hierarchy** from Zombie to NecromancerKing, each with unique AI behaviours
- **4 tower types** (Arrow, Magic, Siege, Support) with upgrade paths
- **Tetris-style inventory** where items have 2D sizes (e.g., a sword is 1×3, chest armour is 2×3)
- **Equipment system** with 8 slots (weapon, offhand, helmet, chest, legs, boots, 2 accessories)
- **Item rarity system** (Common → Legendary) with stat multipliers
- **Bone pile mechanic** where dead enemies leave corpses that bosses can resurrect
- **Skill tree system** with branching prerequisites (combat, magic, utility trees)
- **Shop system** with wave-scaled stock and buy/sell mechanics
- **SQLite persistence** for saving and loading game state via JDBC
- **30 hand-crafted waves** with mini-bosses every 5 waves and a final boss at wave 30

Technology Stack

Component	Technology	Version
Language	Java	21 (OpenJDK)
GUI Framework	JavaFX	21.0.2
Database	SQLite via JDBC	sqlite-jdbc 3.45.1.0
Build	Manual javac with sources.txt	—
IDE	VS Code with Java Extension Pack	—

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2. System Architecture Overview

JavaTower follows a **layered architecture** with clear separation of concerns:

```
┌──────────────────────────────────────────────────────────┐
│                    Presentation Layer                    │
│   GameGUI · GameBoard · HeroPanel · ShopPanel · etc.     │
├──────────────────────────────────────────────────────────┤
│                    Game Logic Layer                      │
│   WaveManager · Shop · Inventory · SkillTree · Combat    │
├──────────────────────────────────────────────────────────┤
│                    Entity Layer                          │
│   Entity → Hero, Enemy (10 subclasses), Tower (4 sub.)   │
│   Item · BonePile                                        │
├──────────────────────────────────────────────────────────┤
│                    Factory Layer                         │
│   EnemyFactory · TowerFactory · ItemFactory              │
├──────────────────────────────────────────────────────────┤
│                    Data / Persistence Layer              │
│   DatabaseManager (SQLite / JDBC)                        │
```

```
├──────────────────────────────────────────────────────────┤
│                    Utilities                              │
│   Constants · GameState                                   │
└──────────────────────────────────────────────────────────┘
```

Data flow: The `GameGUI` (presentation) orchestrates the game loop. Each frame, it calls `update()` on entities (Hero, Enemies, Towers), which read from `Constants` and interact via the systems layer. Factories create new entities. The `DatabaseManager` persists state to SQLite.

---

3. Architecture Diagram

```
                        ┌──────────┐
                        │  Main    │
                        │ (Entry)  │
                        └──────────┘
                             │
                             │ launches
                             ▼
                   ┌──────────────────┐
                   │      GameGUI     │ ◄── AnimationTimer (60 FPS)
                   │   (Application)  │
                   └──┬──┬──┬──┬──┬──┘
                      │  │  │  │  │
        ┌─────────────┘  │  │  │  └────────────────┐
        ▼                ▼  │  ▼                   ▼
┌──────────┐  ┌──────────┐│┌──────────┐  ┌──────────────┐
│ GameBoard│  │ HeroPanel│└│ ShopPanel │  │InventoryPanel│
│ (Canvas) │  │          │ │          │  │              │
└──────────┘  └──────────┘ └──────────┘  └──────────────┘
                             │
              ┌──────────────┼──────────────┐
              ▼              ▼              ▼
         ┌──────────┐   ┌──────────┐    ┌──────────┐
         │   Hero   │   │  Enemies │    │  Towers  │
         │(Entity)  │   │ (Entity) │    │ (Entity) │
         └──────────┘   └──────────┘    └──────────┘
              │              │              │
              ▼              ▼              ▼
┌──────────────┐ ┌──────────────┐ ┌──────────────┐
│  Inventory   │ │ WaveManager  │ │ TowerFactory │
│  Shop        │ │ EnemyFactory │ │              │
│  SkillTree   │ │              │ │              │
└──────────────┘ └──────────────┘ └──────────────┘
        │
        ▼
┌────────────────┐
│ DatabaseManager │ ◄── SQLite via JDBC
│   (Singleton)  │
└────────────────┘
```

---

4. Package Structure

```
javatower/
├── util/                          # Configuration and state management
│   ├── Constants.java             # Game-wide constants (dimensions, speeds, ranges)
│   └── GameState.java             # State machine enum (MAIN_MENU, PLAYING, etc.)
│
├── entities/                      # Core game entities (OOP hierarchy)
│   ├── Entity.java                # Abstract base class for all entities
│   ├── Hero.java                  # Player character (extends Entity)
│   ├── Enemy.java                 # Abstract enemy base (extends Entity)
│   ├── Tower.java                 # Abstract tower base (extends Entity)
│   ├── Item.java                  # Equipment and consumable items
│   ├── BonePile.java              # Dead enemy corpses for resurrection
│   ├── enemies/                   # Concrete enemy implementations
│   │   ├── Zombie.java            # Tier 1 — slow melee
│   │   ├── Skeleton.java          # Tier 2 — ranged
│   │   ├── Ghoul.java             # Tier 3 — fast melee
│   │   ├── Wight.java             # Tier 4 — standard melee
│   │   ├── Wraith.java            # Tier 5 — phasing ranged
│   │   ├── Revenant.java          # Tier 6 — resurrects on death
│   │   ├── DeathKnight.java       # Tier 7 — heavy melee
│   │   ├── Lich.java              # Tier 8 — kiting summoner
│   │   ├── BoneColossus.java      # Tier 9 — massive tank
│   │   └── NecromancerKing.java   # Tier 10 — final boss, dual-weapon AI
│   └── towers/                    # Concrete tower implementations
│       ├── ArrowTower.java        # Single-target physical damage
│       ├── MagicTower.java        # AoE magic damage
│       ├── SiegeTower.java        # Slow, high-damage siege
│       └── SupportTower.java      # Buff/support aura
│
├── systems/                       # Game systems and mechanics
│   ├── WaveManager.java           # Wave progression and enemy spawning
│   ├── Shop.java                  # Buy/sell items between waves
│   ├── Inventory.java             # Tetris-style grid inventory
│   ├── SkillTree.java             # Branching skill progression
│   ├── SkillNode.java             # Individual skill node
│   └── CombatSystem.java          # Combat coordination (legacy)
│
├── factories/                     # Object creation (Factory Pattern)
│   ├── EnemyFactory.java          # Creates enemies with wave scaling
│   ├── TowerFactory.java          # Creates towers by type
│   └── ItemFactory.java           # Creates items and shop stock
│
├── gui/                           # JavaFX user interface
│   ├── GameGUI.java               # Main application window and game loop
│   ├── GameBoard.java             # Canvas-based game rendering
│   ├── HeroPanel.java             # Hero stats display
│   ├── WaveInfoPanel.java         # Wave progress display
│   ├── ShopPanel.java             # Shop interface
│   ├── InventoryPanel.java        # Inventory management interface
```

```
│   └── SkillTreePanel.java        # Skill tree interface
│
└── database/                      # Data persistence
    └── DatabaseManager.java       # SQLite JDBC singleton
```

Total: 35 Java source files across 7 packages.

---

5. Class Definitions

5.1 Core Entities (`javatower.entities`)

`Entity` (Abstract Base Class)

The root of the entity hierarchy. All game objects that exist on the game board extend this class.

Attribute	Type	Description
`name`	`String`	Entity display name
`maxHealth`	`int`	Maximum hit points
`currentHealth`	`int`	Current hit points
`attack`	`int`	Base attack damage
`defence`	`int`	Damage reduction
`x, y`	`double`	Pixel position (continuous for smooth movement)
`radius`	`double`	Collision radius in pixels
`alive`	`boolean`	Whether the entity is active

Method	Description
`takeTurn()`	Abstract — called each frame (overridden by subclasses)
`takeDamage(int)`	Applies damage reduced by defence (minimum 1); triggers `onDeath()`
`heal(int)`	Restores health up to maximum
`setPosition(double, double)`	Sets pixel coordinates
`distanceTo(Entity)`	Euclidean distance to another entity
`distanceTo(double, double)`	Euclidean distance to a point
`overlaps(Entity)`	Circle-based collision detection
`getHealthPercent()`	Returns health as percentage (for health bars)
`onDeath()`	Hook method called when health reaches zero

OOP Principle: Abstraction — `Entity` defines a common interface for all game objects while leaving specific behaviour to subclasses.

`Hero` (extends `Entity`)

The player-controlled character with equipment, inventory, skills, and real-time AI.

Attribute	Type	Description
`weapon`, `offhand`, `helmet`, `chest`, `legs`, `boots`, `accessory1`, `accessory2`	`Item`	8 equipment slots
`level`	`int`	Current level (starts at 1)
`experience`	`int`	Current XP
`experienceToNextLevel`	`int`	XP threshold (scales per level)
`gold`	`int`	Currency for shop purchases
`mana`, `maxMana`	`int`	Resource for abilities
`critChance`	`int`	Critical hit percentage
`skillPoints`	`int`	Points for unlocking skills
`moveSpeed`	`double`	Pixels per second
`attackCooldown`	`double`	Seconds between attacks
`attackTimer`	`double`	Time accumulator
`targetX`, `targetY`	`double`	Click-to-move destination
`moving`	`boolean`	Whether hero is moving
`inventory`	`Inventory`	Tetris grid inventory (3×3)
`combatTree`, `magicTree`, `utilityTree`	`SkillTree`	Three skill branches

Key Method	Description
`update(double dt, List<Enemy>)`	Per-frame update: smooth movement + auto-attack nearest enemy
`moveTo(double, double)`	Sets click-to-move target
`attackEnemy(Enemy)`	Deals damage with critical hit chance
`gainExperience(int)`	Awards XP and checks for level-up
`levelUp()`	Increases stats and grants skill point
`equipItem(Item)`	Equips item to the correct slot based on `Item.Slot`
`unequipItem(Item.Slot)`	Removes equipped item back to inventory

OOP Principle: Encapsulation — all hero state is private with controlled access through methods.

`Enemy` (Abstract, extends `Entity`)

Base class for all undead enemies with a built-in type system for 10 tiers.

Inner Enum — `EnemyType`:

Type	Tier	HP	ATK	DEF	XP	Gold	Special
ZOMBIE	1	30	5	2	10	5	Slow melee
SKELETON	2	25	8	1	15	8	Ranged (150px)
GHOUL	3	40	7	3	20	12	Fast melee
WIGHT	4	60	10	5	30	18	Standard
WRAITH	5	35	12	2	25	15	Phasing, ranged (80px)
REVENANT	6	80	12	8	40	25	Self-resurrect, mini-boss
DEATH_KNIGHT	7	120	15	12	60	40	Heavy melee
LICH	8	70	20	6	80	50	Kiting AI, summoner
BONE_COLOSSUS	9	200	18	15	100	70	Massive tank
NECROMANCER_KING	10	500	30	20	500	200	Final boss, dual-weapon AI

Attribute	Type	Description
`type`	`EnemyType`	The enemy's tier and base stats
`speed`	`double`	Movement speed (pixels/second)
`attackCooldown`	`double`	Seconds between attacks
`attackRange`	`double`	Attack reach in pixels
`isBoss`, `isMiniBoss`	`boolean`	Boss flags
`canPhase`, `canResurrect`, `canSummon`	`boolean`	Special ability flags
`siblings`	`List<Enemy>`	Sibling reference for pack behaviour
`bonePiles`	`List<BonePile>`	Reference for summoning mechanics

Key Method	Description
`update(double, Hero, List<Enemy>)`	Abstract real-time AI update
`smoothMoveToward(double, double, double, double)`	Movement with collision avoidance
`summonFromBones(List<BonePile>)`	Hook for summoner enemies
`onDeath()`	Hook for death effects (resurrection, etc.)

OOP Principle: Inheritance and Polymorphism — 10 concrete subclasses override `update()` to implement unique AI behaviours.

`Tower` (Abstract, extends `Entity`)

Base class for defensive structures placed on the game board.

Attribute	Type	Description
`type`	`TowerType`	ARROW, MAGIC, SIEGE, or SUPPORT
`range`	`int`	Abstract range units (×64 = pixels)
`damage`	`int`	Base damage per attack
`attackCooldown`	`double`	Seconds between attacks
`upgradeLevel`	`int`	Current upgrade tier
`upgradeCost`	`int`	Gold cost for next upgrade

Key Method	Description
`update(double, List<Enemy>)`	Per-frame: attacks enemies on cooldown
`getRangePixels()`	Converts range to pixel distance
`attack(List<Enemy>)`	Abstract — deals damage to enemy/enemies
`selectTarget(List<Enemy>)`	Abstract — targeting strategy
`upgrade()`	Abstract — improves tower stats

`Item`

Represents equipment and consumable items with a Tetris-style 2D size system.

Inner Enums:

- `Slot` : WEAPON, OFFHAND, HELMET, CHEST, LEGS, BOOTS, ACCESSORY, CONSUMABLE
- `Rarity` : COMMON (×1.0), UNCOMMON (×1.2), RARE (×1.5), EPIC (×2.0), LEGENDARY (×3.0)

Attribute	Type	Description
`name`, `description`	`String`	Item identity
`slot`	`Slot`	Equipment slot category
`rarity`	`Rarity`	Rarity with stat multiplier
`width`, `height`	`int`	2D inventory size
`statBonuses`	`Map<String, Integer>`	Stat modifications (attack, defence, critChance, etc.)
`itemLevel`	`int`	Level scaling factor
`buyPrice`, `sellPrice`	`int`	Shop prices

Static Factory Methods: `createSword()` , `createShield()` , `createHelmet()` , `createChestArmor()` , `createRing()` , `createHealthPotion()` , `createManaPotion()`

Item Sizes (Tetris Grid):

Item	Width × Height	Slot
Sword	1 × 3	WEAPON
Shield	2 × 2	OFFHAND
Helmet	2 × 1	HELMET
Chest Armor	2 × 3	CHEST
Ring	1 × 1	ACCESSORY
Health/Mana Potion	1 × 1	CONSUMABLE

`BonePile`

Represents a dead enemy's corpse on the game board. Used by summoner enemies (Lich, NecromancerKing) to resurrect fallen undead.

Attribute	Type	Description
`x, y`	`double`	Position on the game board
`enemyType`	`EnemyType`	Type of fallen enemy
`tier`	`int`	Tier of fallen enemy
`used`	`boolean`	Whether this pile has been consumed

5.2 Enemy Subclasses (`javatower.entities.enemies`)

Each enemy overrides `update()` with unique AI behaviour:

Class	Tier	AI Behaviour
`Zombie`	1	Direct melee chase — slow, relentless
`Skeleton`	2	Ranged attacker — maintains distance, shoots from 150px
`Ghoul`	3	Fast melee — rushes hero, high speed
`Wight`	4	Standard melee — balanced stats
`Wraith`	5	Phasing ranged — phases through obstacles, ranged attack at 80px
`Revenant`	6	Self-resurrect — comes back once after death, mini-boss
`DeathKnight`	7	Heavy melee — slow but extremely durable
`Lich`	8	Kiting summoner — maintains 200px distance, flees within 100px, summons from bone piles, ranged attack at 1.8s cooldown
`BoneColossus`	9	Massive tank — very slow, enormous HP and defence
`NecromancerKing`	10	**Final boss** — dual-weapon system (sword melee 0–50px at 1.5× damage, bow ranged 80–260px at 0.7× damage), dead zone 50–80px, kiting AI, summoning,

		enrage at 25% HP

OOP Principle: Polymorphism — all enemies share the `Enemy` interface but each exhibits radically different behaviour when `update()` is called.

5.3 Tower Subclasses (`javatower.entities.towers`)

Class	Type	Behaviour
`ArrowTower`	ARROW	Targets nearest enemy, single-target physical damage, fast fire rate
`MagicTower`	MAGIC	Area-of-effect magic damage, targets clusters
`SiegeTower`	SIEGE	Slow fire rate, massive single-target damage, long range
`SupportTower`	SUPPORT	Buffs nearby towers and hero with aura effects

Each implements `attack()`, `selectTarget()`, and `upgrade()` with distinct strategies.

5.4 Systems (`javatower.systems`)

`Inventory`

A **Tetris-style** grid inventory where items occupy multiple cells based on their `width × height`.

Attribute	Type	Description
`width, height`	`int`	Grid dimensions (starts 3×3)
`occupied`	`boolean[][]`	Cell occupancy map
`itemGrid`	`Item[][]`	Maps cells to items
`items`	`List<Item>`	Fast item list

Key Method	Description
`canPlaceItem(Item, int, int)`	Checks if an item fits at position
`addItem(Item)`	Auto-finds first available placement
`addItemAt(Item, int, int)`	Places item at specific grid position
`removeItem(int, int)`	Removes item occupying the given cell
`expand()`	Increases grid size (progression reward)

`WaveManager`

Manages the progression through 30 waves with boss encounters.

Key Method	Description
`startWave()`	Spawns enemies for current wave, flags boss waves

`spawnEnemies()`	Delegates to `EnemyFactory` for wave composition
`onEnemyKilled(Enemy)`	Tracks kill count for wave completion
`isWaveComplete()`	Returns true when all enemies defeated
`advanceWave()`	Increments wave counter

`Shop`

Handles buying and selling items between waves.

Key Method	Description
`refreshStock(int)`	Generates new items via `ItemFactory` scaled to wave level
`buyItem(Hero, Item)`	Purchases item if hero has enough gold
`sellItem(Hero, Item)`	Removes item and refunds gold
`getAffordableItems(int)`	Filters shop stock by hero's gold

`SkillTree` and `SkillNode`

A branching skill progression system with three trees (combat, magic, utility).

`SkillNode` attributes: `id`, `name`, `description`, `branch`, `cost`, `prerequisites` (list of node IDs), `bonuses` (stat map), `specialEffect` (Runnable).

`SkillTree` manages a list of nodes and validates prerequisite chains before unlocking.

`CombatSystem`

Legacy class retained for compatibility. Real-time combat is now handled directly by `Hero.update()`, `Enemy.update()`, and `Tower.update()` within the `AnimationTimer` game loop.

5.5 Factories (`javatower.factories`)

All factories follow the **Factory Method** design pattern.

`EnemyFactory`

Static Method	Description
`createWaveEnemies(int wave)`	Returns hand-crafted enemy list for waves 1–29
`createMiniBoss(int wave)`	Creates a mini-boss for boss waves (5, 10, 15, 20, 25)
`createFinalBoss()`	Creates the NecromancerKing for wave 30

Enemy compositions are hand-designed per wave (not random), ensuring difficulty progression:

- Waves 1–4: Basic enemies (Zombie, Skeleton, Ghoul)
- Waves 5–9: Mid-tier with mini-boss (Wight, Wraith)
- Waves 10–14: Advanced enemies (Revenant, DeathKnight)
- Waves 15–24: Elite encounters (Lich, BoneColossus)

- Waves 25–29: Mixed elite waves
- Wave 30: Final boss — NecromancerKing

`TowerFactory`

Creates tower instances by `TowerType` with appropriate stats and costs.

`ItemFactory`

Static Method	Description
`generateShopStock(int wave, int count)`	Creates random items scaled to wave level
`createRandomItem(int level)`	Generates a random equipment piece with rarity roll

5.6 GUI (`javatower.gui`)

`GameGUI` (extends `Application`)

The main JavaFX application. Manages:

- **Scene management:** Main menu, game view, shop overlay, inventory overlay, skill tree overlay
- **Game loop:** `AnimationTimer` with delta-time calculation
- **Input handling:** Click-to-move, tower placement mode
- **State transitions:** `GameState` enum drives which UI is active
- **Wave progression:** Automatic wave advancement with delay timer

`GameBoard` (extends `Canvas`)

Pixel-based rendering of the game world:

- Grid lines, bone piles (skull icon for tier ≥ 5), tower range circles
- Entity rendering with health bars and tier-based colour coding
- Click handlers for hero movement and tower placement

UI Panels

Panel	Purpose
`HeroPanel`	Displays hero stats (HP, ATK, DEF, level, gold, mana)
`WaveInfoPanel`	Shows current wave, enemies remaining
`ShopPanel`	Buy/sell interface between waves
`InventoryPanel`	Item management and equipment
`SkillTreePanel`	Skill tree interaction

5.7 Database (`javatower.database`)

`DatabaseManager` (Singleton)

Provides SQLite persistence via JDBC for save/load functionality.

Tables:

Table	Purpose	Key Columns
`save_state`	Current game save	hero stats, position, wave number, timestamp
`meta_progression`	Cross-run unlocks	unlocked items, skill unlocks, max wave reached

Key Method	Description
`getInstance()`	Returns the singleton instance
`initialize()`	Creates tables if not exist
`saveGame(Hero, int)`	Persists hero state and wave number
`loadGame()`	Restores hero from saved state

OOP Principle: Singleton pattern ensures a single database connection throughout the application lifecycle.

5.8 Utilities (`javatower.util`)

`Constants`

Centralises all magic numbers as `public static final` constants:

Category	Constants
Screen	`SCREEN_WIDTH` (960), `SCREEN_HEIGHT` (640)
Grid	`TILE_COLS` (15), `TILE_ROWS` (10), `TILE_SIZE` (64px)
Entity Sizes	`HERO_RADIUS` (20px), `ENEMY_RADIUS_SMALL/MEDIUM/LARGE/BOSS`
Speeds	`HERO_SPEED` (120 px/s), `ENEMY_SPEED_SLOW/NORMAL/FAST`
Combat	`MELEE_RANGE` (40px), `TOWER_RANGE_UNIT` (64px)
Economy	`STARTING_GOLD` (50), `MAX_WAVES` (30), `WAVE_DELAY` (3.0s)

`GameState` (Enum)

State machine for the application: `MAIN_MENU`, `PLAYING`, `SHOPPING`, `SKILL_TREE`, `PAUSED`, `GAME_OVER`, `VICTORY`.

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6. OOP Principles Applied

Encapsulation

- All entity fields are `private` with getter/setter access
- `Inventory` hides the 2D grid algorithms behind simple `addItem()` / `removeItem()` methods
- `DatabaseManager` encapsulates all SQL operations behind clean Java methods
- `Hero` equipment managed internally — external code only calls `equipItem()`

Inheritance

- `Entity` → `Hero`, `Enemy`, `Tower` hierarchy provides shared state and behaviour

- `Enemy` → 10 concrete subclasses, each specialising AI behaviour
- `Tower` → 4 concrete subclasses with different attack strategies

Inheritance Hierarchy:

```
Entity (abstract)
├── Hero
├── Enemy (abstract)
│   ├── Zombie
│   ├── Skeleton
│   ├── Ghoul
│   ├── Wight
│   ├── Wraith
│   ├── Revenant
│   ├── DeathKnight
│   ├── Lich
│   ├── BoneColossus
│   └── NecromancerKing
└── Tower (abstract)
    ├── ArrowTower
    ├── MagicTower
    ├── SiegeTower
    └── SupportTower
```

Polymorphism

- `Enemy.update()` — each subclass implements unique AI (kiting, summoning, phasing, resurrecting)
- `Tower.attack()` / `Tower.selectTarget()` — each tower type has a different targeting and damage strategy
- `Entity.takeDamage()` — can be overridden by subclasses for special damage handling
- `Entity.onDeath()` — hook method overridden for bone pile creation, resurrection, etc.

Abstraction

- `Entity` defines what all game objects *can do* without specifying *how*
- `Tower.attack()` and `Tower.selectTarget()` are abstract, forcing subclasses to define behaviour
- `GameState` enum abstracts application states into a finite state machine
- Factory classes abstract the creation complexity behind simple static methods

---

7. Design Patterns

Factory Method Pattern

Used in: `EnemyFactory`, `TowerFactory`, `ItemFactory`

Centralises object creation, allowing wave-specific enemy compositions and randomised items without exposing construction logic to the caller.

```
// Client code doesn't need to know which concrete Enemy to create
List<Enemy> wave = EnemyFactory.createWaveEnemies(currentWave);
```

Singleton Pattern

Used in: `DatabaseManager`

Ensures a single database connection is shared across the entire application:

```
DatabaseManager.getInstance().saveGame(hero, waveNumber);
```

Template Method Pattern

Used in: `Entity.takeDamage()` → `onDeath()`

The `takeDamage()` method defines the algorithm (reduce HP, check death), and subclasses hook into `onDeath()` for custom behaviour (bone pile spawning, resurrection).

State Pattern (Implicit)

Used in: `GameState` enum + `GameGUI`

The application behaviour changes based on `GameState`: `PLAYING` runs the game loop, `SHOPPING` pauses gameplay and shows the shop, etc.

Strategy Pattern (Implicit)

Used in: `Tower` subclasses

Each tower implements a different targeting and attack strategy through polymorphic `selectTarget()` and `attack()` methods.

Observer Pattern (Implicit)

Used in: `WaveManager.onEnemyKilled()`

Game systems are notified of enemy deaths to update wave progress, spawn bone piles, and award XP/gold.

---

8. Class Diagram (UML)

```
┌──────────────────────────────────────────────────────────┐
│                      <<abstract>>                        │
│                        Entity                            │
├──────────────────────────────────────────────────────────┤
│ - name: String                                           │
│ - maxHealth, currentHealth: int                          │
│ - attack, defence: int                                   │
│ - x, y: double                                           │
│ - radius: double                                         │
│ - alive: boolean                                         │
├──────────────────────────────────────────────────────────┤
│ + takeTurn(): void {abstract}                            │
│ + takeDamage(int): int                                   │
│ + heal(int): void                                        │
│ + distanceTo(Entity): double                             │
│ + overlaps(Entity): boolean                              │
│ + onDeath(): void                                        │
└──────────────┬──────────────┬──────────────┬────────────┘
```

```
            |                |                |
  ┌─────────▼──┐    ┌───────▼──────┐   ┌─────▼──────────┐
  |   Hero     |    |   <<abs>>    |   |   <<abs>>      |
  |            |    |   Enemy      |   |   Tower        |
  ├────────────┤    ├──────────────┤   ├────────────────┤
  | -weapon    |    | -type        |   | -type          |
  | -offhand   |    | -speed       |   | -range         |
  | -helmet    |    | -attackRange|   | -damage        |
  | -chest     |    | -isBoss      |   | -cooldown      |
  | -legs      |    | -canSummon   |   | -upgradeL.     |
  | -boots     |    | -canPhase    |   ├────────────────┤
  | -acc1,2    |    | -bonePiles   |   | +attack()      |
  | -level     |    ├──────────────┤   | +selectT.()    |
  | -gold      |    | +update()    |   | +upgrade()     |
  | -inventory|    | +summon()    |   └────────┬───────┘
  | -skillTree|    └──────┬───────┘           |
  ├────────────┤           |           ┌──────┴───────┐
  | +update()  |     ┌─────┴──────┐    |ArrowTower    |
  | +moveTo()  |     | Zombie     |    |MagicTower    |
  | +equip()   |     | Skeleton   |    |SiegeTower    |
  | +levelUp()|     | Ghoul      |    |SupportTower|
  └────────────┘     | Wight      |    └──────────────┘
        |            | Wraith     |
        |            | Revenant   |
  ┌─────▼──────┐     | DeathKn.   |
  |  Inventory |     | Lich       |
  ├────────────┤     | BoneColo.  |
  | -width,height|   | NecroKing  |
  | -occupied[][] |  └────────────┘
  | -itemGrid[][] |
  ├────────────┤
  | +addItem()  |        ┌──────────┐
  | +removeItem()|       |   Item   |
  | +canPlace() |        ├──────────┤
  | +expand()   |        | -slot    |
  └────────────┘         | -rarity  |
                         | -width,h |
                         | -statMap |
                         | -level   |
                         └──────────┘
┌──────────────┐
|DatabaseManager |
|  <<singleton>> |
├──────────────┤         ┌──────────┐
| -instance    |         | BonePile |
| -connection  |         ├──────────┤
├──────────────┤         | -x, y    |
| +getInstance() |       | -enemyType|
| +saveGame()  |         | -used    |
| +loadGame()  |         └──────────┘
└──────────────┘
```

9. Database Design

JavaTower uses **SQLite** via **JDBC** (`sqlite-jdbc-3.45.1.0.jar`) for persistent storage.

Entity-Relationship

```
┌──────────────────────────────┐      ┌──────────────────────────────┐
│         save_state           │      │      meta_progression        │
├──────────────────────────────┤      ├──────────────────────────────┤
│ id (PK)        INTEGER       │      │ id (PK)        INTEGER       │
│ hero_name      TEXT          │      │ unlocked_items TEXT           │
│ hero_level     INT           │      │ skill_unlocks  TEXT           │
│ hero_hp        INT           │      │ max_wave       INT            │
│ hero_max_hp    INT           │      │ total_gold     INT            │
│ hero_attack    INT           │      │ inventory_width INT           │
│ hero_defence   INT           │      │ inventory_height INT          │
│ hero_gold      INT           │      └──────────────────────────────┘
│ hero_mana      INT           │
│ hero_max_mana  INT           │
│ hero_exp       INT           │
│ hero_x         INT           │
│ hero_y         INT           │
│ wave           INT           │
│ timestamp      DATETIME      │
└──────────────────────────────┘
```

JDBC Connection Example

```java
// Singleton initialisation
connection = DriverManager.getConnection("jdbc:sqlite:javatower.db");

// Parameterised query (prevents SQL injection)
PreparedStatement ps = connection.prepareStatement(
    "INSERT OR REPLACE INTO save_state (id, hero_name, ...) VALUES (1, ?, ?, ...)");
ps.setString(1, hero.getName());
ps.setInt(2, hero.getLevel());
ps.executeUpdate();
```

---

10. GUI Architecture

The JavaFX GUI is built around a `BorderPane` layout:

```
┌──────────────────────────────────────────────────────┐
│                  WaveInfoPanel (Top)                 │
├──────────────────────────────────┬──────────────────┤
│                                  │                  │
│          GameBoard               │   HeroPanel      │
│         (Canvas)                 │   (Right)        │
│     960 × 640 pixels             │                  │
```

```
│                                  │  - HP bar        │
│   Renders:                       │  - Stats         │
│   - Grid lines                   │  - Level/XP      │
│   - Hero (blue circle)           │  - Gold          │
│   - Enemies (colour by tier)     │  - Equipment     │
│   - Towers (coloured squares)    │                  │
│   - Bone piles (gray/skull)      │                  │
│   - Health bars                  │                  │
│   - Tower range indicators       │                  │
│                                  │                  │
├──────────────────────────────────┴──────────────────┤
│               Tower Buttons (Bottom)                 │
│          [Arrow] [Magic] [Siege] [Support]           │
│            [Shop] [Inventory] [Skills]               │
└──────────────────────────────────────────────────────┘
```

Overlay Screens (pause game loop when open):

- `ShopPanel` — displayed between waves, buy/sell items
- `InventoryPanel` — manage equipment and items
- `SkillTreePanel` — spend skill points on upgrades

---

11. Game Loop and Real-Time Systems

The game uses JavaFX `AnimationTimer` for a real-time game loop:

```java
gameLoop = new AnimationTimer() {
    @Override
    public void handle(long now) {
        double dt = (now - lastNanoTime) / 1_000_000_000.0;  // Delta time in seconds
        lastNanoTime = now;

        hero.update(dt, enemies);           // Hero movement & auto-attack
        for (Enemy e : enemies)
            e.update(dt, hero, enemies);    // Enemy AI
        for (Tower t : towers)
            t.update(dt, enemies);          // Tower auto-attack

        checkDeaths();                      // Clean up dead entities, spawn bone piles
        checkWaveCompletion();              // Advance waves
        gameBoard.render(hero, enemies,     // Redraw everything
                         towers, bonePiles);
    }
};
```

Delta-time system ensures consistent gameplay speed regardless of frame rate:

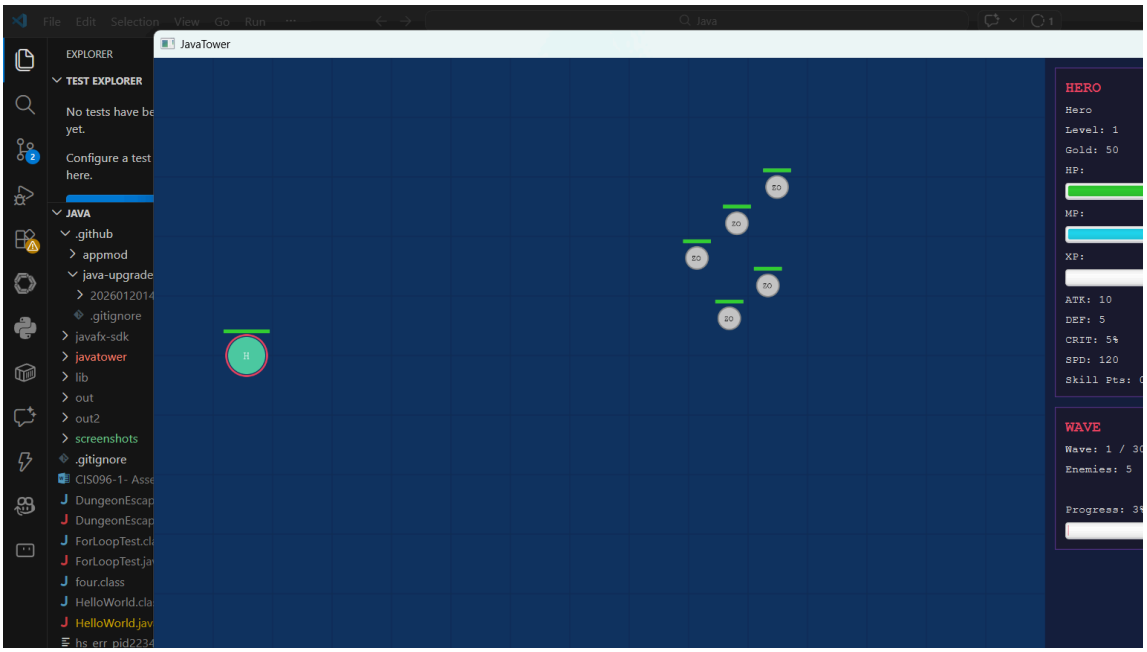
- Movement: `newPos = oldPos + speed * dt`
- Attack cooldowns: `timer += dt; if (timer >= cooldown) { attack(); timer = 0; }`

---

12. Application Screenshots

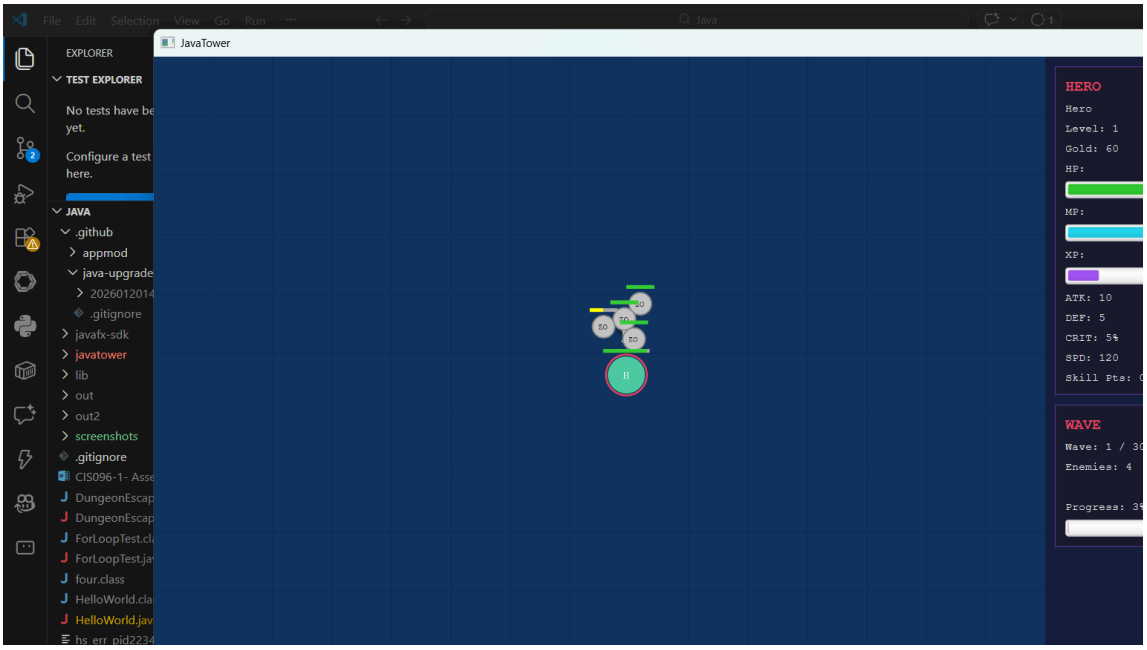
The following screenshots demonstrate the working application:

Main Menu



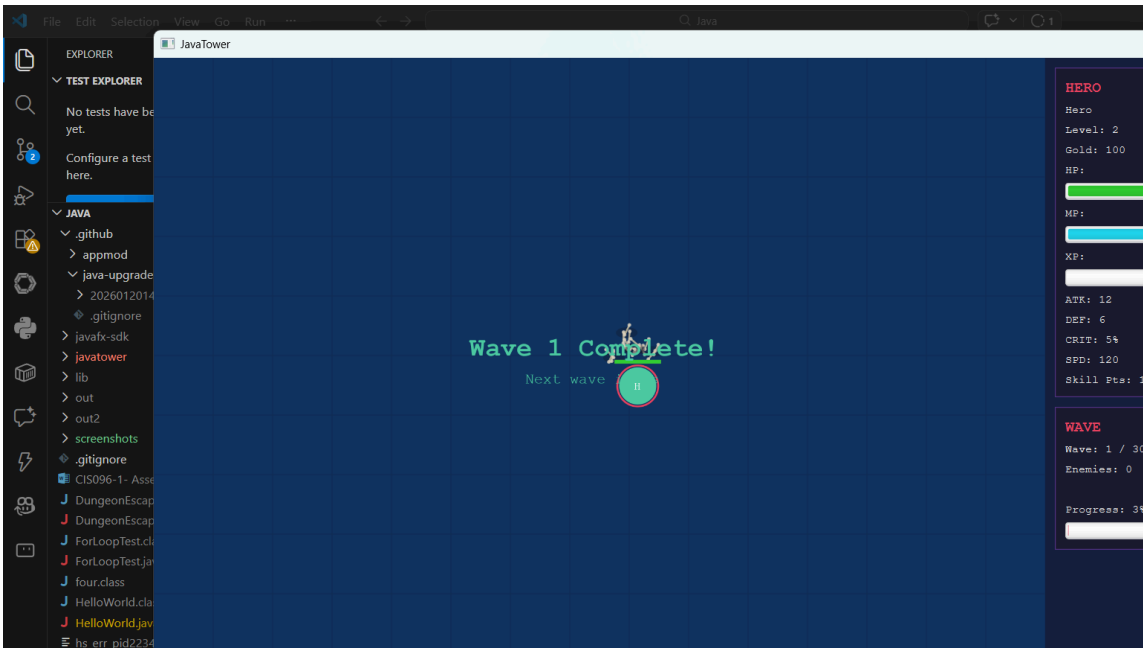
The main menu screen with New Game, Load Game, and Quit options.

Gameplay — Wave Start



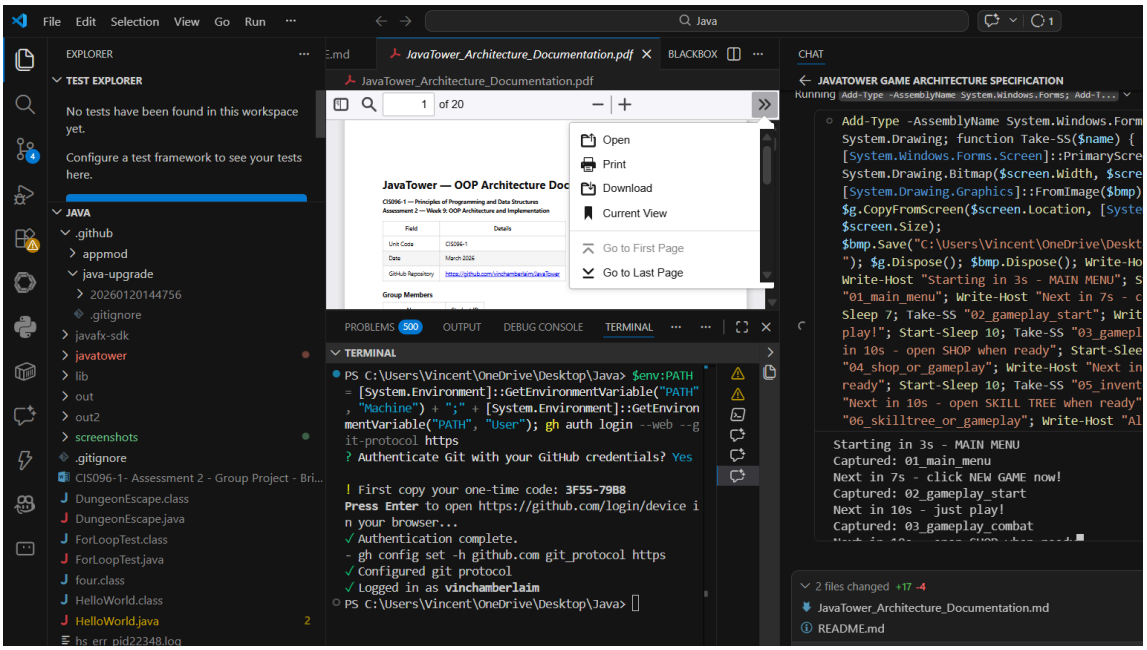
The game board showing the hero, grid, tower placement buttons, and HUD panels.

Gameplay — Combat



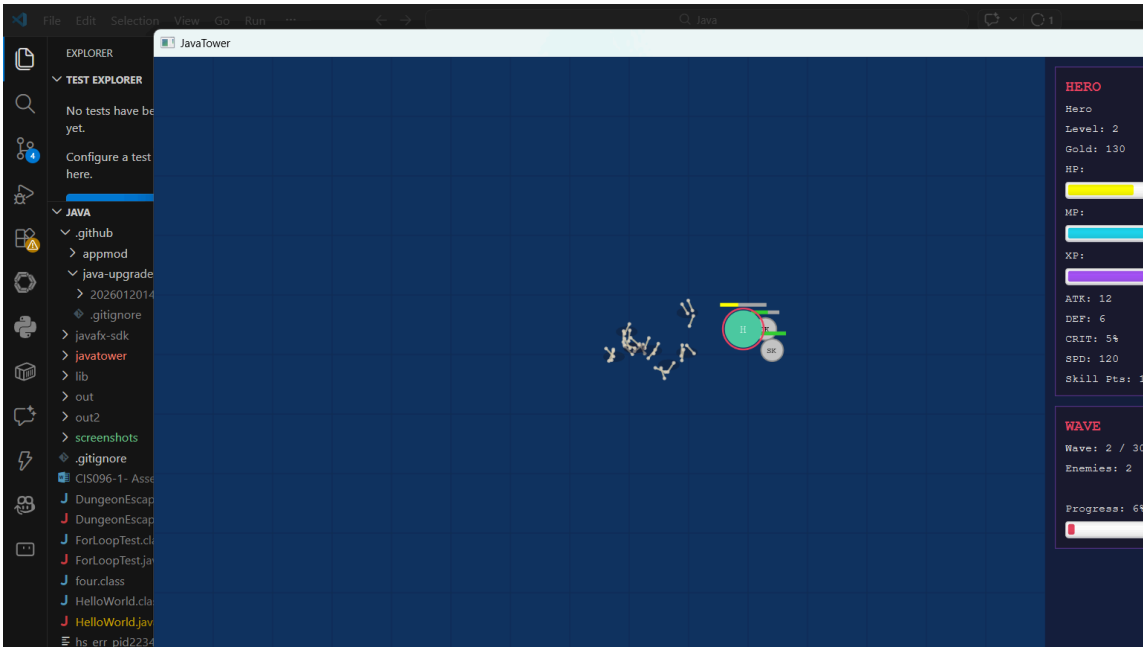
Real-time combat with enemies approaching the hero, health bars visible.

Shop / Gameplay



The shop interface or continued gameplay showing wave progression.

Inventory



The inventory management screen or gameplay with towers placed.

Skill Tree / Additional Gameplay

The skill tree interface or late-game combat.

---

13. Testing and Demonstration

Functional Testing

The following core features have been tested through gameplay demonstration:

Feature	Test	Status
Hero movement	Click-to-move across the game board	✓ Working
Hero auto-attack	Hero attacks nearest enemy in melee range	✓ Working
Enemy AI	Each enemy type exhibits unique behaviour	✓ Working
Tower placement	Click tower button, then click grid cell	✓ Working
Tower attacks	Towers auto-attack enemies within range	✓ Working
Wave progression	30 waves with auto-advance	✓ Working
Boss encounters	Mini-bosses at waves 5/10/15/20/25, final boss at 30	✓ Working
Bone pile system	Dead enemies leave corpses; summoners resurrect them	✓ Working
Shop buy/sell	Purchase and sell items between waves	✓ Working
Inventory grid	Tetris-style item placement and removal	✓ Working
Equipment system	8 equipment slots with item types	✓ Working
Level/XP system	Hero gains XP from kills, levels up with stat increases	✓ Working
Database save/load	SQLite persistence of hero state	✓ Working
NecromancerKing AI	Dual-weapon kiting boss with summoning and enrage	✓ Working
Lich kiting AI	Maintains distance, flees when too close, summons	✓ Working

Compilation Verification

The project compiles cleanly with zero errors:

```
javac --module-path "javafx-sdk/lib" --add-modules javafx.controls,javafx.graphics \
      -cp "lib/sqlite-jdbc-3.45.1.0.jar" -d out2 @sources.txt
```

Runtime Verification

The application launches and runs all 30 waves without crashes:

```
java --module-path "javafx-sdk/lib" --add-modules javafx.controls,javafx.graphics \
     -cp "out2;lib/sqlite-jdbc-3.45.1.0.jar" Main
```

---

14. AI Tool Usage and Reflection

Tools Used

- **GitHub Copilot** — Assisted with code generation, debugging, and architecture documentation
- **VS Code** — Primary IDE with Java Extension Pack

How AI Helped

- **Code scaffolding:** Copilot generated boilerplate for entity classes and factory methods, saving significant time on repetitive code
- **Debugging:** AI helped identify issues such as the `double` to `int` cast in `DatabaseManager.saveGame()` for hero position coordinates
- **Architecture planning:** AI assisted in structuring the package hierarchy and identifying appropriate design patterns
- **Documentation:** AI helped create comprehensive class descriptions and diagrams

Challenges Without AI

- Understanding the interaction between JavaFX's `AnimationTimer` and real-time game logic would have required more trial and error
- Designing the Tetris-style inventory algorithm (2D bin packing) was conceptually challenging
- Balancing 10 enemy tiers with unique AI behaviors across 30 waves would have taken significantly longer

What I Learned

- How to structure a large Java project with proper package separation
- How to implement real-time game loops with delta-time
- How `abstract` classes and `polymorphism` enable extensible enemy/tower systems
- How JDBC connects Java applications to SQLite databases
- The importance of the Factory pattern for managing complex object creation

---

15. Conclusion

JavaTower demonstrates comprehensive application of Object-Oriented Programming principles through a real-time tower defence RPG with 35 Java source files across 7 packages. The project features:

- A deep **inheritance hierarchy** (Entity → Hero/Enemy/Tower with 14 subclasses)
- Effective use of **polymorphism** for diverse enemy AI and tower behaviour
- Strong **encapsulation** across all classes with private fields and controlled access
- Proper **abstraction** through abstract base classes and interfaces
- **Design patterns** including Factory Method, Singleton, Template Method, and State
- **JDBC database connectivity** for persistent game saves
- A **JavaFX GUI** with real-time canvas rendering and multiple interactive panels

The codebase is organised, documented, and fully functional — compiling and running cleanly on Java 21 with JavaFX 21.0.2 and SQLite JDBC 3.45.1.0.

This document was prepared for the CIS096-1 Week 9 submission — OOP Architecture and Implementation.